EXP.NO:20

Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

PROBLEM STATEMENT:

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

CODE

```c
#include<stdio.h>

int main(){
    int n;
    scanf("%d", &n);

    int arr[n];

    for(int i=0;i<n;i++)
        scanf("%d", &arr[i]);

    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(arr[i]==arr[j]){
                printf("%d", arr[i]);
                return 0;
            }
        }
    }
    return 0;
}
```

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OUTPUT:



	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.